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Experience

Postdoctoral Research Fellow, Arnaout Lab, Beth Israel Deaconess Medical Center, 2022-2026.
Supervisor: Dr. Ramy Arnaout

Postdoctoral Research Fellow, Computer Science Department, Boston College, 2021-2022.
Supervisor: Prof. José Bento

Education

Ph.D. Physics, University of Texas at Austin, May 2021.
Thesis advisor: Prof. Willy Fischler
GPA: 3.60

B.S. Physics and B.S. Mathematics, Oklahoma State University, August 2013.

Honors and Awards

OGS Summer Only Fellowship Summer 2018
Nominated for UT Services for Students with Disabilities Appreciation Award Fall 2016

Publications and Preprints

P. Nguyen, J. Couch, R. Bansul, A. Morgan, C. Tam, M. Li, R. Arnaout, R. Arnaout, "Which Similarity-Sensitive Entropy?," [arXiv:2511.03849 [cs.IT]].

J. Couch, M. Li, R. Arnaout, R. Arnaout, "X-Factor: Quality Is a Dataset-Intrinsic Property," [arXiv:2505.22813 [cs.LG]].

J. Couch, R. Arnaout, R. Arnaout, "Beyond Size and Class Balance: Alpha as a New Dataset Quality Metric for Deep Learning," [arXiv:2407.15724 [cs.CV]].

J. Couch, R. Arora, J. Braun, J. Kaplinsky and E. Hill, A. Li, B. Altshul, R. Arnaout, "Scaling Monte-Carlo-Based Inference on Antibody and TCR Repertoires," [arXiv:2312.12525 [q-bio.QM]].

J. Couch, P. Nguyen, S. Racz, G. Stratis and Y. Zhang, "Possibility of entanglement of purification to be less than half of the reflected entropy," Phys. Rev. A **109**, no.2, 022426 (2024) doi:10.1103/PhysRevA.109.022426 [arXiv:2309.02506 [quant-ph]]. 3 citations counted in INSPIRE as of 04 Oct 2025

J. Couch, Y. Fan and S. Shashi, "Circuit Complexity in Topological Quantum Field Theory," *Fortsch. Phys.* **70**, no.9-10, 2200102 (2022) doi:10.1002/prop.202200102 [arXiv:2108.13427 [hep-th]]. 17 citations counted in INSPIRE as of 04 Oct 2025

E. Caceres, S. Chapman, J. D. Couch, J. P. Hernandez, R. C. Myers and S. M. Ruan, "Complexity of Mixed States in QFT and Holography," *JHEP* **03**, 012 (2020) doi:10.1007/JHEP03(2020)012 [arXiv:1909.10557 [hep-th]]. 106 citations counted in INSPIRE as of 04 Oct 2025

J. Couch, S. Eccles, P. Nguyen, B. Swingle and S. Xu, "Speed of quantum information spreading in chaotic systems," *Phys. Rev. B* **102**, no.4, 045114 (2020) doi:10.1103/PhysRevB.102.045114 [arXiv:1908.06993 [cond-mat.stat-mech]]. 41 citations counted in INSPIRE as of 04 Oct 2025

E. Cáceres, J. Couch, S. Eccles and W. Fischler, "Holographic Purification Complexity," *Phys. Rev. D* **99**, no.8, 086016 (2019) doi:10.1103/PhysRevD.99.086016 [arXiv:1811.10650 [hep-th]]. 34 citations counted in INSPIRE as of 04 Oct 2025

J. Couch, S. Eccles, T. Jacobson and P. Nguyen, "Holographic Complexity and Volume," *JHEP* **11**, 044 (2018) doi:10.1007/JHEP11(2018)044 [arXiv:1807.02186 [hep-th]]. 64 citations counted in INSPIRE as of 04 Oct 2025

J. Couch, S. Eccles, W. Fischler and M. L. Xiao, "Holographic complexity and noncommutative gauge theory," *JHEP* **03**, 108 (2018) doi:10.1007/JHEP03(2018)108 [arXiv:1710.07833 [hep-th]]. 78 citations counted in INSPIRE as of 04 Oct 2025

J. Couch, W. Fischler and P. H. Nguyen, "Noether charge, black hole volume, and complexity," *JHEP* **03**, 119 (2017) doi:10.1007/JHEP03(2017)119 [arXiv:1610.02038 [hep-th]]. 269 citations counted in INSPIRE as of 04 Oct 2025

Conferences and Workshops

IAIFI summer workshop 2024, IAIFI, Cambridge, MA. - Speaker

Mexicuerdas 2019, Instituto de Ciencias Nucleares, UNAM, Mexico City, Mexico - Speaker

Southwest Strings Meeting 2019, Arizona State University, Tempe, AZ. - Speaker

Joint meeting of the Texas Section of the APS, Texas Section of the AAPT, and Zone 13 of the SPS 2018, University of Houston, Houston, TX - Speaker

Prospects in Theoretical Physics 2018, "From Qubits to Spacetime", Institute for Advanced Study, Princeton, NJ - Participant

Southwest Holography Meeting 2018, University of Texas at Austin, Austin, TX. - Speaker

Joint meeting of the Texas Section of the APS, Texas Section of the AAPT, and Zone 13 of the SPS 2017, University of Texas at Dallas, Richardson, TX - Speaker

"Quantum Information in Quantum Gravity III" (2017), University of British Columbia, Vancouver, BC - Participant

It from Qubit "Complexity and Black Holes" Workshop (2017), Stanford Institute for Theoretical Physics, Stanford University, Stanford, CA - Participant

Talks

"Diversity with Similarity as a Measure of Training Set Quality," IAIFI summer workshop 2024, MIT, Cambridge, MA August 2024

"The Speed of Quantum Information Spreading in Chaotic Systems," Mexicuerdas 2019, Instituto de Ciencias Nucleares, UNAM, Mexico City, Mexico December 2019

"Holographic Complexity and Volume," Southwest Strings Meeting 2019, Arizonal State University, Tempe, AZ February 2019

"Purification complexity in free scalar field theory," Joint Meeting of the Texas Section of the APS, Texas Section of the AAPT, and Zone 13 of the SPS, Houston, TX October 2018

"Holographic Complexity," Oklahoma State Universtiy, Stillwater, OK April 2018

"Holographic Complexity and Non-Commutative Guage Theory," Southwest Holography Meeting 2018, University of Texas at Austin, Austin, TX March 2018

"Holographic Complexity and Non-Commutative Guage Theory," Joint Meeting of the Texas Section of the APS, Texas Section of the AAPT, and Zone 13 of the SPS, Richardson, TX October 2017

Teaching Experience

PHY 373 (Quantum Physics I: Foundations) - Grader and Discussion TA Fall 2015 & 2018, Spring & Fall 2020

PHY 341 (Forensic Science) - Grader Fall 2017 and 2019

PHY 389K (Graduate Quantum Mechanics I) - Grader Spring 2019

PHY 362L (Quantum Physics III: Particles and Nuclei) - Grader Spring 2016 & 2019

PHY 316 (Electricity and Magnetism) - Grader and Discussion TA Spring 2017 & 2018

PHY 352K (Classical Electrodynamics I) - Grader and Discussion TA Fall 2016

PHY 103M (Lab for Engineering Physics II) - Teaching Assistant S/F 2014 & S/Summer 2015

PHY 103N (Lab for Engineering Physics I) - Teaching Assistant Fall 2013